



NORTH SYDNEY BOYS HIGH SCHOOL

MATHEMATICS

2010 Year 7 Mathematics Assessment Task 1

General instructions

- Working time – 50 minutes.
- Write your answers in the space provided.
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- Calculators are **NOT** to be used under any circumstances.
- All necessary working should be shown in every question.
- Attempt **all** questions.

Class (please ✓)

- ☐ 7N – Mr Lam
- ☐ 7R – Mr Berry
- ☐ 7T – Mr Lowe/Mr Trenwith
- ☐ 7H – Mr Ireland/Ms O'Brien
- ☐ 7S – Mr Trenwith

NAME:

Marker's use only.

SECTION	1	2	3	4	Total	%
MARKS	$\overline{29}$	$\overline{23}$	$\overline{16}$	$\overline{6}$	$\overline{74}$	

Section I: Whole numbers

1. Evaluate 8

(a) $823 + 627.$

(b) $9\,765 - 1\,688.$

.....

(c) $97 \times 68.$

(d) $1\,900 \div 75.$

.....

2. What year was this foreign TV program produced? 1



.....

3. (a) Write 97661 in expanded notation. 2

.....

- (b) Write as numerals: *six hundred and sixty five million two hundred and thirteen.* 1

.....

4. Find the sum of three hundred and forty five with the product of nine thousand and sixty and twenty two. 2

.....

5. Round 27 649 to the nearest: 2

(a) ten.

(b) thousand.

.....

.....

6. Calculate: 8

(a) $3 + 72 \div 12$.

(b) $(96 \div 3) + (258 \div 2)$.

.....

.....

.....

.....

(c) $(2^3 + 4)^2$.

(d) $20 \div (9 - 4) + 6$.

.....

.....

.....

.....

7. Insert $<$, $>$ or $=$ to make a true statement: 2

(a) $43 + 7 \square 25 + 25$.

(b) $298 + 546 \square 1\,000 - 154$.

8. An *About Maths 7* textbook costs thirty five dollars to purchase including shipping. Mr Rezcallah needs to replace all one hundred and fifty textbooks. In addition, each textbook case costs six dollars. How much does Mr Rezcallah have to pay to replace all the textbooks and cases? 3

.....

.....

.....

Section II: Sets & Venn Diagrams

9. Given $n(A)$ is the number of elements in the set A , find $n(A)$ if 4

(a) $A = \{0, 1, 2, \dots, 10\}.$

(b) $A = \emptyset.$

.....

.....

(c) $A = \{\text{primes less than } 20\}.$

(d) $A = \{0, 1, 1\}.$

.....

.....

10. Choose $\subset$ or $\not\subset$ to make a true statement: 4

(a) $\{\text{animals}\} \square \{\text{mammals}\}$

(b) $\{\text{primes}\} \square \{\text{odd numbers}\}$

(c) $\{\text{adults}\} \square \{\text{people}\}.$

(d) $\emptyset \square (\{\text{odd numbers}\} \cap \{\text{even numbers}\}).$

11. If $A = \{1, 2, 3, 4, \dots, 10\}$, $B = \{1, 2, 3, 8, 9\}$ and $C = \{2, 3, 5, 7, 9\}$, 6

(a) Find $A \cap B.$

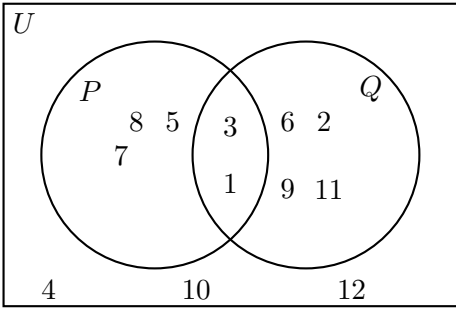
(b) Find $B \cup C.$

.....

.....

(c) Draw the Venn diagram that represents the above information.

12. From the Venn diagram below, use set notation to list the following sets: 6



- (a) $P \cap Q$.

.....
- (b) $P \cup Q$.

.....
- (c) $\overline{P}$.

.....
- (d) $\overline{Q}$.

.....
- (e) $\overline{(P \cup Q)}$.

.....
- (f) $P \cap \overline{Q}$.

.....

13. A coin collector buys a bag containing 100 coins. Sixty-five are silver, 90 are round, and 61 are silver and round. Draw a Venn diagram to find out how many are neither silver nor round. 3

.....

Examination continued overleaf...

Section III: Number Theory

- 14.** List all the factors of: **4**

(a) 80

(b) 144

.....

.....

.....

.....

.....

.....

- 15.** Write 384 as a product of its prime factors, leaving it in index form. **2**

.....

- 16.** (a) Draw a factor tree for 48 and another for 64. **2**

(b) For the numbers 48 and 64, find the

2

i. Highest common factor

ii. Lowest common multiple

.....

.....

.....

.....

Examination continued overleaf...

17. By considering prime factors, calculate

4

(a) $\sqrt{2304}$.

(b) $\sqrt[3]{3375}$.

.....

18. Is the number 123 456 789 divisible by 9? Explain your answer.

2

.....

.....

.....

Examination continued overleaf...

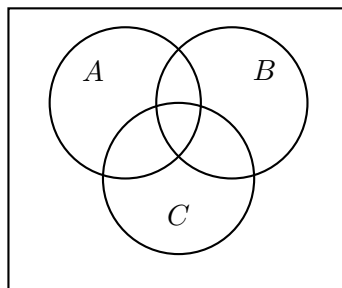
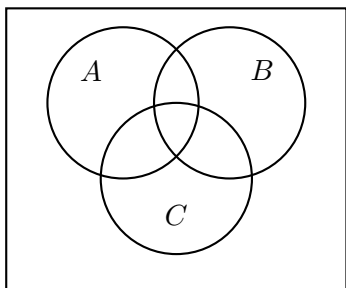
Section IV: Working Mathematically

19. Shade in the part of the Venn diagram that is represented by:

4

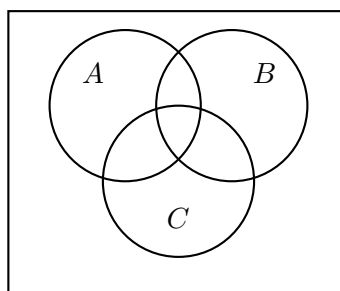
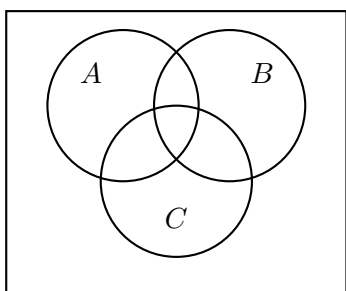
(a) $A \cap B \cap C$.

(b) $(A \cap B) \cup C$.



(c) $A \cap C \cap \overline{B}$.

(d) $(A \cup B \cup C) \cap \overline{(A \cap B \cap C)}$.



20. A bus route takes 75 min to complete a journey in one direction. If there is to be a bus running every 10 minutes in each direction and the driver of each bus must take 10 minutes rest between starting a trip in the opposite direction, how many buses are required to operate the route in both directions?

2

.....

.....

.....

End of examination

Marking scheme note: where there are less marks allocated to a question than the answers (e.g. factors of 80, 2 marks), award full marks for no errors and subtract one mark for each mistake.

Section I: Whole Numbers

1. (a) (2 marks)

$$\begin{array}{r} 1 \\ 823 \\ + 627 \\ \hline 1450 \end{array}$$

- (b) (2 marks)

$$\begin{array}{r} 9765 \\ - 1688 \\ \hline 8077 \end{array}$$

- (c) (2 marks)

$$\begin{array}{r} 97 \\ \times 68 \\ \hline 776 \\ 582 \\ \hline 6596 \end{array}$$

- (d) (2 marks)

$$\begin{array}{r} 25 \\ 75 \overline{)1900} \\ \underline{1500} \\ 400 \\ \underline{375} \\ 25 \end{array}$$

2. (1 mark)

$$\text{MCMXCIII} = 1993$$

3. (a) (2 marks)

$$(9 \times 10^4) + (7 \times 10^3) + (6 \times 10^2) + (6 \times 10^1) + 1$$

Award full marks even if not in “scientific notation”, i.e. $9 \times 10\,000$.

- (b) (1 mark)

$$665\,000\,213$$

(Note that the wording is ambiguous and may be 11 880 345).

4. (2 marks)

$$199\,665$$

$$\begin{array}{r} 9060 \\ \times 22 \\ \hline 18120 \\ 18120 \\ \hline 199320 \end{array} \quad \begin{array}{r} 199320 \\ + 345 \\ \hline 199665 \end{array}$$

5. (a) (1 mark)

$$27\,650$$

- (b) (1 mark)

$$28\,000$$

6. (a) (2 marks)

$$3 + (72 \div 12) = 3 + 6 = 9$$

- (b) (2 marks)

$$(96 \div 3) + (258 \div 2) = 32 + 129 = 161$$

- (c) (2 marks)

$$(2^3 + 4)^2 = (8 + 4)^2 = 144$$

- (d) (2 marks)

$$20 \div (9 - 4) + 6 = (20 \div 5) + 6 = 10$$

7. (a) (1 mark) “=”

- (b) (1 mark) “<”

$$\begin{array}{r} 11 \\ 298 \\ + 546 \\ \hline 844 \end{array} \quad \begin{array}{r} 1000 \\ - 154 \\ \hline 846 \end{array}$$

8. (3 marks)

$$\$ (35 + 6) \times 150$$

$$\begin{array}{r} 41 \\ \times 150 \\ \hline 2050 \\ 41 \\ \hline 6150 \end{array}$$

Section II: Sets & Venn Diagrams

9. (a) (1 mark)

$$n(A) = 11$$

(b) (1 mark)

$$n(\emptyset) = 0$$

(c) (1 mark)

$$A = \{2, 3, 5, 7, 11, 13, 17, 19\}$$

$$n(A) = 8$$

(d) (1 mark)

$$n(A) = 2$$

10. (a) (1 mark) $\not\subset$.(b) (1 mark) $\not\subset$.(c) (1 mark) $\subset$.(d) (1 mark) $\subset$.

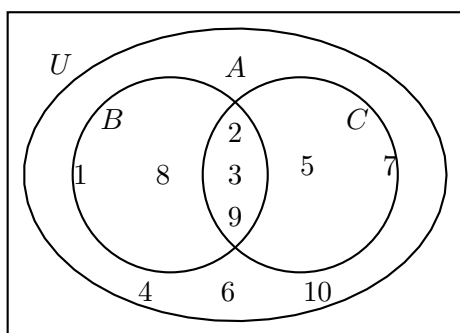
11. (a) (2 marks)

$$A \cap B = \{1, 2, 3, 8, 9\}$$

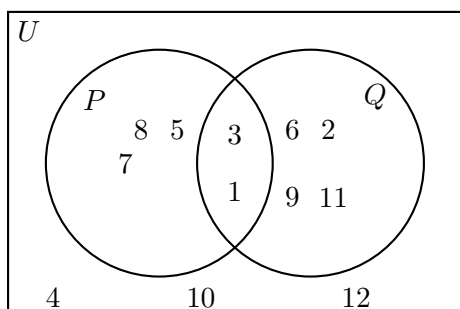
(b) (2 marks)

$$B \cup C = \{1, 2, 3, 5, 7, 8, 9\}$$

(c) (2 marks)



12.



(a) (1 mark)

$$P \cap Q = \{1, 3\}$$

(b) (1 mark)

$$P \cup Q = \{1, 2, 3, 5, 6, 7, 8, 9, 11\}$$

(c) (1 mark)

$$\overline{P} = \{2, 4, 6, 9, 10, 11, 12\}$$

(d) (1 mark)

$$\overline{Q} = \{4, 5, 7, 8, 10, 12\}$$

(e) (1 mark)

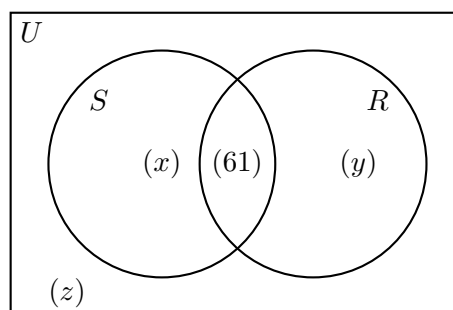
$$\overline{P \cup Q} = \{4, 10, 12\}$$

(f) (1 mark)

$$P \cap \overline{Q} = \{5, 7, 8\}$$

13. (3 marks)

- ✓ [1] for correct Venn diagram
- ✓ [1] for finding x and y (number of silver/round coins)
- ✓ [1] for correct final answer.



$$x + 61 = 65$$

$$x = 4$$

Hence there are 4 silver coins.

$$y + 61 = 90$$

$$y = 29$$

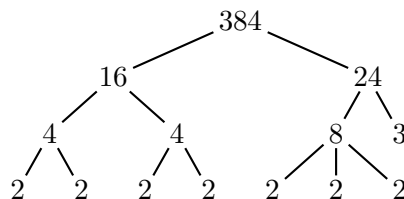
Hence there are 29 round coins and in total, $61 + 29 + 4 = 94$ silver and/or round coins. $\therefore$ there are 6 that are neither silver or round.

Section III: Number Theory**14.** (a) (2 marks)

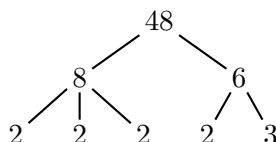
1, 2, 4, 5, 8, 10, 16, 20, 40, 80

(b) (2 marks)

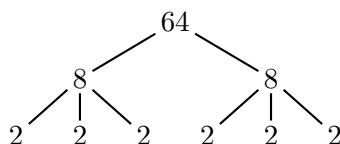
1, 2, 3, 4, 6, 8, 9, 12, 16, 18, 24, 36, 48, 72, 144

15. (2 marks)

$$\therefore 384 = 2^7 \times 3$$

16. (a) (2 marks)

$$48 = 2^4 \times 3$$



$$64 = 2^6$$

(b) i. (1 mark)

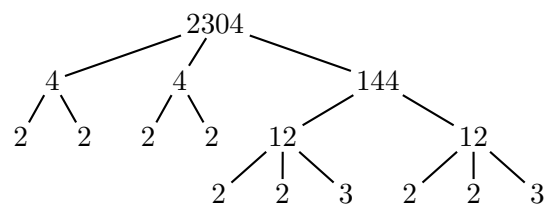
$$\gcd(48, 64) = 2^4 = 16$$

ii. (1 mark)

$$\text{lcm}(48, 64) = 2^6 \times 3 = 192$$

17. (a) (2 marks)

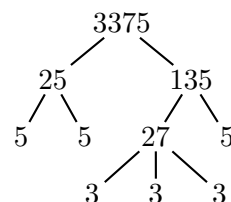
$$\begin{array}{r} 576 \\ 4 \overline{) 2304} \\ \underline{2000} \\ 304 \\ \underline{280} \\ 24 \\ \underline{24} \\ 0 \end{array} \quad \begin{array}{r} 144 \\ 4 \overline{) 576} \\ \underline{400} \\ 176 \\ \underline{160} \\ 16 \\ \underline{16} \\ 0 \end{array}$$



$$\begin{aligned} 2304 &= 2^8 \times 3^2 \\ \therefore \sqrt{2304} &= 2^4 \times 3 = 48 \end{aligned}$$

(b) (2 marks)

$$\begin{array}{r} 135 \\ 25 \overline{) 3375} \\ \underline{2500} \\ 875 \\ \underline{750} \\ 125 \\ \underline{125} \\ 0 \end{array}$$



$$\begin{aligned} 3375 &= 5^3 \times 3^3 \\ \therefore \sqrt{3375} &= 5 \times 3 = 15 \end{aligned}$$

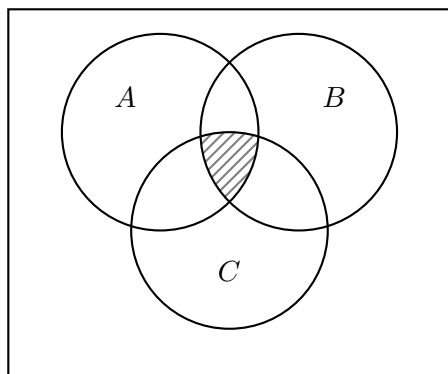
18. (2 marks)

✓ [1] for correct answer (yes)

✓ [1] for correct explanation

$$\begin{aligned} &1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 \\ &= (1 + 8) + (2 + 7) + (3 + 6) + (5 + 4) + 9 \\ &= 5 \times 9 \\ &= 45 \end{aligned}$$

As the sum of its digits is divisible by 9,
the number itself is divisible by 9.

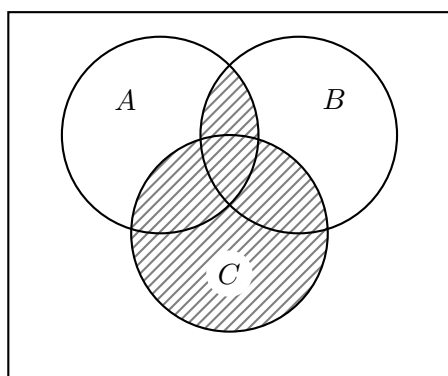
Section IV:**20.** (2 marks)**19.** (a) (1 mark)

- ✓ [1] for correct calculation of total round trip time including rest.
 ✓ [1] for correct final answer.

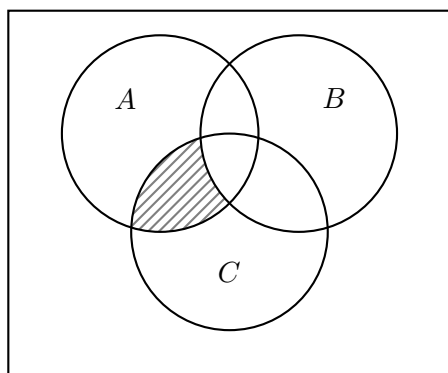
$$(75 \times 2) + (10 \times 2) = 150 + 20 = 170 \text{ min}$$

$$\frac{170 \text{ min}}{10 \text{ min/bus}} = 17 \text{ buses}$$

(b) (1 mark)



(c) (1 mark)



(d) (1 mark)

